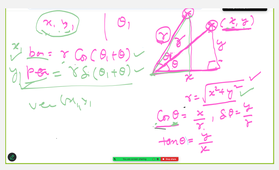


$$x, y \xrightarrow{\theta_1} (x_1, y_1)$$

$$x_1 = r\cos(\theta_1+\theta) = r[\cos\theta_1\cos\theta - \sin\theta_1\sin\theta] = x\cos\theta_1 - y\sin\theta_1$$

$$y_1 = r\sin(\theta_1+\theta) = r[\sin\theta_1\cos\theta + \sin\theta\cos\theta] = x\sin\theta_1 + y\sin\theta_1$$

$$[x, y] \times \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = [x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta]$$

1×2 $\quad 2\times 2$

$x, y = 3, 4$

$\theta = 90$

$x_1 = x\cos 90 - y\sin 90 = -4$

$y_1 = x\sin 90 + y\cos 90 = 3$

$\theta = 90$

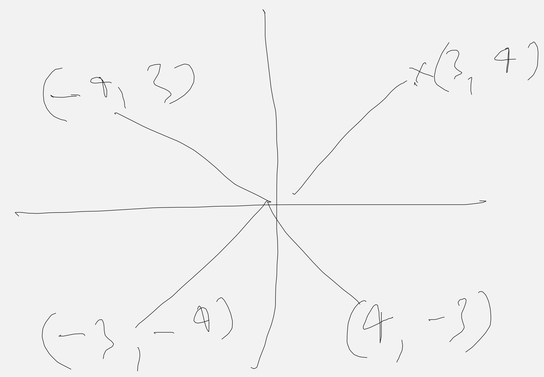
$x_2 = -3$

$y_2 = -4$

$\theta = 90 \xrightarrow{\theta = 90} x_4 = 3$, $y_4 = 4$

$x_3 = 4$

$y_3 = -3$



$2x + y = 5 \longrightarrow [2, 1, 5]$

$x - y = 6 \longrightarrow [1, -1, 6]$

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+2 & 2+2 & 3+2 \\ 3+4 & 6+4 & 9+4 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 5 \\ 7 & 10 & 13 \end{bmatrix} \checkmark$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$3\times 4 \quad 4\times 5$

$$= [3\times 5]$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \times \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$= \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}$$